

Power Play — Full Test

Mixed questions from all four concepts. Section A is interactive; Sections B and C are written. Time: 45 minutes ·
Total: 28 marks. Answers on the separate key.

Objective (1 mark each)**Q1 · MCQ****[1]**

In 2^n , n is known as the:

(a) base (b) exponent (c) factor (d) product

Q2 · MCQ**[1]**

$2^{-3} = ?$

(a) 8 (b) -8 (c) $1/8$ (d) $-1/8$

Q3 · MCQ**[1]**

$3^4 \times 3^2 = ?$

(a) 3^6 (b) 3^8 (c) 9^6 (d) 3^2

Q4 · MCQ**[1]**

$(2/3)^{-2} = ?$

(a) 49 (b) $9/4$ (c) -49 (d) 6

Q5 · FILL IN THE BLANK**[1]**

For any non-zero integer y , the value of y^0 is:

Q6 · TRUE / FALSE**[1]**

$(a^2)^3 = a^5$.

True False

Q7 · MCQ

[1]

540 expressed as prime powers is:

- (a) $2^2 \times 3^3 \times 5$ (b) $2^3 \times 3^2 \times 5$ (c) $2^2 \times 3^2 \times 5^2$ (d) $2^2 \times 3^3$

Q8 · ASSERTION-REASON

[1]

Assertion (A): Any non-zero number raised to the power 0 is 1.

Reason (R): $a^m \div a^m = a^{m-m} = a^0 = 1$.

- (a) Both true, R explains A (b) Both true, R does not explain A (c) A true, R false
(d) A false, R true

Short answer (2 marks each)

Q9

[2]

Simplify, writing each with a positive exponent: (i) $2^{-4} \times 2^7$ (ii) $6^7 \div 6^3$.

Q10

[2]

Evaluate: (i) $(-2/3)^4$ (ii) $1/4^{-2}$.

Q11

[2]

Write in standard form: (i) 78000 (ii) 0.000045.

Q12

[2]

Express 405 and 540 as products of prime powers.

Long answer (3 marks each)

Q13

[3]

Simplify, writing each with a positive exponent: (i) $3^2 \times 3^{-5} \times 3^6$ (ii) $p^3 \times p^{-10}$.

Q14

[3]

(i) Express 16^{-2} as a power with base 2. (ii) Express $27/64$ as a power of a rational number.

Q15

[3]

A bacterium doubles every hour, starting from 1 cell. How many cells are there after 10 hours? Give the answer as a power of 2 and as an ordinary number.

Q16

[3]

Star A is 9.35×10^6 km away and star B is 6.27×10^7 km away. Which is nearer, and why?

Total: 28 marks · Check your work against the **answer key**.